Prediction of Surface Subsidence and Strain over Inclined Seam using Influence Function Method

Project –IV (MI57002) report submitted to

Indian Institute of Technology Kharagpur in

partial fulfilment for the award of the degree of

Dual Degree (B.Tech + M.Tech)

in

Mining Engineering

By

Ayan Ahmed

[21MI31007]

Under the Supervision of

Professor Debasis Deb



Department Of Mining Engineering

Indian Institute of Technology Kharagpur

Spring Semester, 2025-26

May 1, 2026

Declaration

I hereby declare that the work contained in this report titled **"Prediction of Surface Subsidence and Strain over Inclined Seam Using Influence Function Method"** has been carried out by me under the supervision of **Professor Debasis Deb**, Department of Mining Engineering, Indian Institute of Technology Kharagpur.

I further declare that:

(a) The work reported in this thesis is the result of my own independent research and has not been submitted to any other Institute or University for the award of any degree or diploma.

(b) I have abided by the norms and guidelines prescribed in the Ethical Code of Conduct of the Institute.

(c) Wherever I have used materials (data, theoretical formulations, figures, and text) from other sources, I have duly acknowledged them through citations and references.

(d) Permission from the copyright holders has been obtained wherever necessary.

Date: 01.05.2026
Place: IIT Kharagpur
Ayan Ahmed
Ayan Ahmed
Roll No.: 21MI31007



Certificate

This is to certify that the thesis report entitled **"Prediction of Surface Subsidence and Strain over Inclined Seam Using Influence Function Method"** submitted by **Ayan Ahmed (Roll No. 21MI31007)** to the **Indian Institute of Technology Kharagpur** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology & Master of Technology in Mining Engineering**, is a record of bona fide research work carried out by him under my supervision and guidance during the academic year **2025–26**.

To the best of my knowledge, the work reported herein is original and has not been submitted to any other University or Institute for the award of any degree or diploma.

Professor Debasis Deb
Department of Mining Engineering
Indian Institute of Technology Kharagpur
Kharagpur – 721302, India

Date: 01.05.2026
Place: IIT Kharagpur



Abstract

Subsidence prediction plays a crucial role in the safe planning and sustainable execution of underground coal mining operations, particularly in longwall extraction. Conventional empirical and profile-function methods are often limited in their generalizability across varying geological conditions and seam configurations. In this work, a comprehensive study was carried out on the **Influence Function Method (IFM)** and **the Stochastic Influence Function Method (SIFM)** to predict surface subsidence induced by underground coal extraction.

The theoretical framework of subsidence and probabilistic influence functions was developed based on **Chapter 12: Subsidence from *Fundamentals and Applications of Rock Mechanics*** by **Professor Debasis Deb** and **Professor Abhiram Kumar Verma**. These concepts were initially implemented numerically through a custom C++ program capable of computing subsidence at all affected surface points for a flat coal seam, successfully reproducing standard subsidence profiles consistent with established theory.

Building upon this foundation, the study was extended to **inclined seam (<12°)** geometries. A novel mathematical formulation based on the stochastic influence function framework was developed to predict subsidence over inclined seams by incorporating geometric transformation of the influence zone, thereby capturing the inherent asymmetry of subsidence troughs. The proposed model was validated against real longwall subsidence monitoring data, showing strong agreement between predicted and observed results.

In the present work, these theoretical and computational advancements have been further translated into a comprehensive software tool. The developed platform allows users to directly input field dimensions, seam orientation, seam geometry, and geomechanical properties, and subsequently visualize displacement components (Ux, Uy, Uz) and strain components (Ex, Ey) across the surface. Additionally, the software enables customized profile plotting along user-defined lines, facilitating detailed subsidence analysis and interpretation.

This research thus contributes both theoretical and practical advancements by providing a generalized and user-friendly approach for subsidence prediction in flat as well as inclined seams. The developed methodology and software have significant applications in subsidence risk assessment, surface infrastructure protection, and efficient mine planning.



Acknowledgements

I express my deepest sense of gratitude to my thesis supervisor, **Professor Debasis Deb**, Department of Mining Engineering, IIT Kharagpur, for his constant guidance, encouragement, and insightful suggestions throughout this research work. His expertise in rock mechanics and numerical modelling has been invaluable in shaping the direction and outcome of this study.

I am also thankful to all faculty members and staff of the **Department of Mining Engineering**, IIT Kharagpur, for providing academic support and laboratory resources essential for the successful completion of this work. I sincerely acknowledge the assistance received from my fellow research scholars and peers for their constructive discussions and feedback.

Finally, I am grateful to my parents and family for their unwavering support, patience, and motivation during my thesis journey. Their belief in me has been my greatest strength.

LIST OF FIGURES

Chapter 2 — Background and Literature Review

Figure 1 Sphere of influence around a surface grid point due to underground extraction.....4
Figure 2 Annular ring discretisation of the circular influence zone around grid point P.....5
Figure 3 Symmetric subsidence trough for a flat coal seam — equal influence angles β.....8
Figure 4 Asymmetric subsidence trough for an inclined seam — unequal influence path lengths...8
Figure 5 Stochastic Influence Function Method applied to a flat seam geometry.....9
Figure 6 Inclined seam geometry in the original (x-y-z) coordinate system.....9
Figure 7 Transformed coordinate system (x'-y'-z') for inclined seam analysis.....10

Chapter 4 — Case Study: Adriyala Mine

Figure 8 Surface plan of the Adriyala Longwall Panel-1 showing monitoring layout.....15
Figure 9 Discretised computational grid overlaid on the Adriyala panel.....16
Figure 10 Fitted vs. measured subsidence along monitoring line S11–S58 (training dataset).....16
Figure 11 Predicted vs. measured subsidence along monitoring line S61–S99 (testing dataset)..17
Figure 12 Colour-mapped vertical subsidence (Uz) field over the Adriyala panel.....18
Figure 13 Colour-mapped Y-strain (Ey) field over the Adriyala panel.....18
Figure 14 Colour-mapped X-strain (Ex) field over the Adriyala panel.....19

LIST OF TABLES

Chapter 2 — Background and Literature Review

Table 2.1 Summary of common subsidence prediction approaches and their key characteristics…4

Chapter 3 — Computational Implementation

Table 3.1 Input parameters and their descriptions in the developed software platform.....11

Table 3.2 Governing equations used for displacement and strain computation.....13

Chapter 4 — Case Study: Adriyala Mine

Table 4.1 Geomechanical and geometric properties of the Adriyala Longwall Panel-1.....15

Table 4.2 Numerical model input parameters used for the Adriyala mine simulation.....16

Table 4.3 RMSE comparison between predicted and observed subsidence (training & testing)....17

LIST OF ABBREVIATIONS

Methods and Techniques

IFM	— Influence Function Method
SIFM	— Stochastic Influence Function Method
FEM	— Finite Element Method
BEM	— Boundary Element Method
RMSE	— Root Mean Square Error
TOC	— Table of Contents

Variables and Symbols

Ux	— Horizontal displacement in the x-direction (m)
Uy	— Horizontal displacement in the y-direction (m)
Uz	— Vertical displacement / subsidence (m)
Ex	— Horizontal strain in the x-direction
Ey	— Horizontal strain in the y-direction
Rf	— Radius of influence (m)
H	— Depth of seam below ground surface (m)
β	— Angle of draw / influence angle (degrees)
θ	— Half-angle between vertical and edge of influence zone (degrees)
α	— Angle between vertical and line joining element centroid to surface point
Smax	— Maximum subsidence factor (m)
p	— Subsidence factor used as model calibration parameter
a	— Influence parameter: $a = H \tan \beta$
Φ(r)	— Stochastic influence function
dA	— Infinitesimal extracted area element (m^2)
ns	— Unit normal vector to the seam plane (nx, ny, nz)
dx, dy	— Grid resolution in x and y directions (m)

Organisations and Locations

IIT KGP	— Indian Institute of Technology Kharagpur
SCCL	— Singareni Collieries Company Limited
DGMS	— Directorate General of Mines Safety, India
A4	— ISO 216 standard paper size (210 mm × 297 mm)

TABLE OF CONTENTS

Preliminary Pages

Declaration i

Certificate ii

Abstract iii

Acknowledgements iv

List of Figures v

List of Tables vi

List of Abbreviations vii

Chapter 1 Introduction 1

1.1 Overview of Subsidence in Rock Mechanics 1

1.2 The Problem of Mining-Induced Subsidence 1

1.3 Project Motivation and Scope 2

1.4 Thesis Structure 3

Chapter 2 Background and Literature Review 4

2.1 Background of Subsidence 4

2.2 Prediction Approaches 4

2.3 Influence Function Method (IFM) 4

2.4 Influence Functions 5

2.5 Horizontal Displacement and Strain 6

2.6 Subsidence Prediction for Inclined Seam 7

Chapter 3 Computational Implementation 11

3.1 Input Structure 11

3.2 Grid Generation 12

3.3 Subsidence Calculation Routine 12

3.4 Strain Calculation 13

3.5 Output and Visualization 14

3.6 Computational Performance 14

Chapter 4 Case Study: Adriyala Mine 15

4.1 Description of the Adriyala Mine 15

4.2 Input Parameters Used in Numerical Model 16

4.3 Actual Subsidence Data vs Model Predictions 16

4.4 Interpretation of Case Study 19

Results and Discussion 20

Conclusion 22

Learning Outcomes and Future Scope 23

References 25

Chapter 1: Introduction

1.1 Overview of Subsidence in Rock Mechanics

Subsidence generally refers to the downward movement, or "settlement," of the ground surface resulting from the caving of underground rock strata. It is a consequence of activities, both natural and manmade, conducted beneath the ground surface. The two most common manmade causes are the **withdrawal of fluids** (such as groundwater, oil, or gas from pores and fractures) and the **disturbance of the ground by excavation of minerals**. In the case of mineral excavation, which is the focus of this study, underground workings induce changes in the stress, strain, and displacement fields of the adjacent soil and rock masses. When these induced displacements propagate to the surface, **subsidence** occurs.

1.2 The Problem of Mining-Induced Subsidence

The development of subsidence, particularly the trough-type observed over longwall panels, is initiated by the **caving** of the immediate and main roof strata above the excavated zone. This process causes a re-adjustment of the overburden, leading to vertical movement on the surface. The surface manifestation of subsidence takes the form of a **subsidence basin**.

The critical importance of studying and predicting subsidence lies in the potential for damage to surface structures, underground utilities, and the disruption of natural drainage systems. Damage to surface structures is primarily caused by **differential settlement** and the resulting **horizontal strains** (compression and tension) along the subsidence trough. Accurate prediction of the subsidence profile (vertical displacement, slope, curvature, and horizontal strain) is essential for effective surface protection and mine planning.

1.3 Project Motivation and Scope

Traditional methods for predicting mining-induced subsidence, such as the **Influence Function Method**, have been developed and successfully applied, often within the context of **flat seams**. However, mining operations, particularly in coalfields, frequently encounter **inclined seams**, where the geometrical and mechanical conditions differ significantly from those in flat deposits. The dip of the seam introduces complexities in the propagation of

displacement, strain, and stress fields, making the direct application of flat-seam models potentially inaccurate.

This project is motivated by the need to develop a robust prediction tool for subsidence over **inclined seams**. It seeks to extend the theoretical foundation of the Influence Function Method by developing a novel **mathematical model** to accurately predict the full subsidence profile and its components over inclined longwall panels.

1.4 Thesis Structure

This thesis consists of six parts:

- **Chapter 1 – Introduction**
  Presents the background of the work, motivation, research gaps and objectives.
- **Chapter 2 – Background and Literature Review**
  Provides fundamentals of subsidence, prediction approaches, influence function method, and extension of the concept to inclined seams (<12°).
- **Chapter 3 – Computational Implementation**
  Describes C++ computational engine and software platform for subsidence, displacement, strain calculation, input handling, and visualization with user-defined profile plotting.
- **Chapter 4 – Case Study**
  Applies the developed model to the Adriyala mine and compares predicted subsidence with actual monitored data.
- **Results and Discussion**
  Interprets the results, explains accuracy and deviations between actual and predicted subsidence.
- **Conclusion and Learning Outcomes &Future Scope and References** Summarises major findings, learning outcomes, and suggests extensions of the research such as modelling steep seams and multiple-seam extraction.

Chapter 2: Background and Literature Review

2.1 Background of Subsidence

Subsidence refers to the downward displacement of the Earth's surface caused by underground mining activities. It is one of the most common environmental impacts of coal and metalliferous mining. The magnitude and distribution of subsidence depend on various factors such as the depth of the seam, geometry of extraction, rock mass properties, and the layout of the mine workings. Predicting subsidence accurately is critical to safeguard surface structures, reduce hazards, and optimize mine design.

2.2 Prediction Approaches

Several approaches have been developed to predict mining-induced subsidence:

- **Empirical methods:** Based on field observations and regional case histories.
- **Analytical methods:** Derive subsidence using mathematical models, such as the **Influence Function Method (IFM)**.
- **Numerical methods:** Apply finite element, boundary element, or hybrid models for detailed simulations.

Among these, the **Influence Function Method** is widely used because of its balance between simplicity and accuracy.

2.3 Influence Function Method (IFM)

Influence function method utilizes the basic theory of superposition to determine the overall influence of an infinitesimal extraction area. Excavation opening is composed of

finite number of small openings or elements. Each small opening produces an individual surface subsidence profile given by an influence function and its summation is the total subsidence. Since the overall subsidence is the summation of influence of each mined out elements, it is not necessary to have contiguous excavation like longwall panel. In other words, this method is well suited for determining surface subsidence caused due to bord and pillar panel.

It is based on the theory that an area of influence develops around a point on the surface due to extraction of a small element within the sphere of influence as shown in Figure 1. The influence is the maximum when element is directly below the surface point being considered. It becomes the minimum when the element is near the edge.

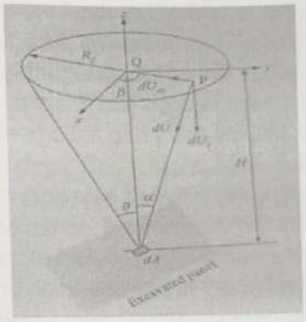


Figure 1

A typical influence function would look at annular zones around a point and use equations to relate the percentage extraction in each area to the subsidence of that point. The area of influence is generally assumed to be circular in shape, although hexagonal shapes have been employed occasionally. The functions commonly used can be subdivided as follows:

(i) Empirically derived functions which use arbitrary constants to produce the desired value and form of subsidence

(ii) Functions relating to the rock properties and mechanics of the problem rather than the arbitrary constants

Second of these groupings is more popular but it is less flexible when carrying out calibration work.

The advantage of this method is the ability to predict subsidence movement at any point above extraction of any practical shape. Drawback of these method is that it is more difficult to apply than profile functions and much more difficult to check and calibrate.

2.4 Influence Functions

The detailed procedure of influence function method is aptly available in various literatures (Peng. 1992; Singh, 1984; Whittaker and Reddish, 1989). The method consists of laying a suitable square or rectangular grid on the surface covering (i) the extraction panel area, and (ii) extending up to the subsidence draw limits (subsidence basin). An influence area of circular shape at each grid point (middle of the grid) is divided into a sufficient number of concentric rings. Each ring is again subdivided into a sufficient number of equal sectors by drawing radii from the grid point (Figure 2). Let us assume a grid centre is denoted by P and hence, the radius of the influence circle is to be taken as Rf = H tan θ, where θ is half angle between the vertical line passing through the grid point and the line connecting the grid point and end of the influence area. The sector which falls inside the excavation area will give rise to subsidence at the grid point P, depending on the weight factor of that sector, and the sector which falls outside the excavation gives no subsidence at P.

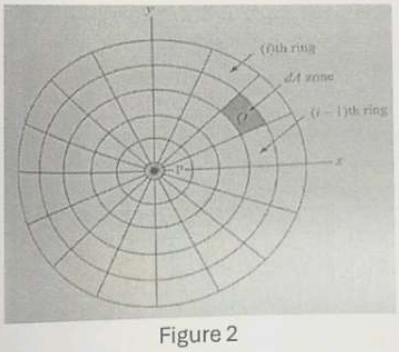


Figure 2

In order to determine weight factor of a sector, first an influence function suitable to the mine site needs to be assumed. There are many kinds of influence functions; they are either derived empirically or are based on theoretical assumptions. Following is the used influence function:

(i) Stochastic influence function:

$$\Phi(r) = \frac{1}{R_f^2} \exp\left(-\frac{\pi r^2}{R_f^2}\right)$$

Equation 1

2.5 Horizontal Displacement and Strain

Let us consider Figure 1 and 2 again. Assume that a vertical subsidence has occurred at grid point P due to extraction of an element dA in underground. The centroid of dA projected on the surface marks the point Q. Let us assume that the element dA lies inside any of the sectors of the influenced circle of grid point P. Then the displacement vector dU at grid point P has two components viz. (i) vertical component dUz obtained by the weight factor, and (ii) horizontal component dUxy in x - y plane measured as given by Equation 2.

$$dUxy = dUz \tan \alpha$$

Equation 2

where α is the angle between the vertical and the line joining the element centroid to the grid point P on the surface. It can be measured as:

$$\tan \alpha = PQ / H$$

Equation 3

It may be noted that if the circular area fully encompasses the extraction panel, then horizontal displacement at grid point P will be zero. The total horizontal movement at point P is again the sum of all extracted elements as Uxy = Σ_N dUxy where N = number of sectors in the influenced zone of grid point P. The resultant dUxy can be resolved in x and y directions as:

$$dU_x = dU_{xy} \cos \beta$$
Equation 4a

$$dU_y = dU_{xy} \sin \beta$$
Equation 4b

where β is the angle between dun and x-axis shown in Figure 1 and can be estimated as $\cos \beta = (Xq - Xp)/PQ$. The total displacement at P is again a sum of all extracted elements. The slope and strain at point P are then readily calculated having known vertical and horizontal movements.

2.6 How to get Subsidence for Inclined Seam?

The classical Influence Function Method, as discussed in the previous sections, assumes a **uniform mining depth** across the entire panel and therefore yields a **symmetric subsidence trough** for flat seams. However, in the case of an inclined coal seam, the extraction depth is no longer constant—surface points located above the up-dip side of the seam are influenced by a shallower excavation depth, while points above the down-dip side correspond to deeper extraction. This spatial variation in depth directly alters the influence radius and intensity of strata disturbance transmitted to the surface, leading to an **asymmetric subsidence trough** that cannot be captured by the standard flat-seam formulation. To extend the mathematical model to inclined seams, the present work retains the **same influence function** and **same theoretical basis**, but introduces a transformation that expresses the **seam depth as a function of position along the dip direction.** By modifying only the geometric input—rather than replacing the influence function itself—the subsidence computation remains fully compatible with the original SIFM framework, while accurately accounting for the changing depth along the seam inclination. This approach enables the prediction of subsidence over both flat and inclined seams within a single unified modelling structure.

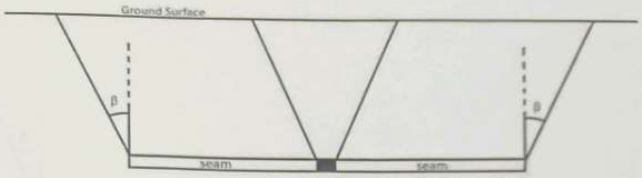


Figure 3

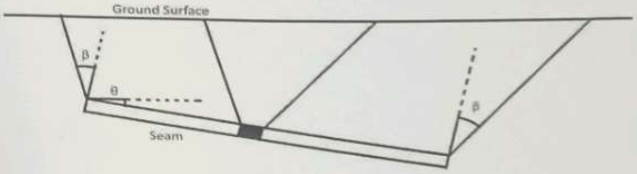


Figure 4

In the conventional Influence Function Method for flat seams, the overburden is assumed to subside symmetrically about the extracted panel because the **depth from the seam to the surface remains constant across the entire panel**. As illustrated in Figure 3, the influence lines originating from the seam boundaries rise vertically and intersect the surface at identical angles β, resulting in a symmetrical subsidence trough and a centrally located peak settlement. However, when the coal seam is inclined, the **depth varies along the dip direction**, and consequently the angles at which strata movement propagates toward the surface become unequal. Figure 4 demonstrates this condition: on the up-dip side the seam lies closer to the surface, causing a shorter influence path and a steeper convergence, while on the down-dip side the greater depth leads to a wider influence spread before reaching the surface. This geometric imbalance shifts the zone of maximum subsidence toward the shallower side and produces a skewed settlement profile, thereby invalidating the symmetric subsidence assumption of the flat-seam model.

Figure 5 below explains stochastics influence function method in flat seam.

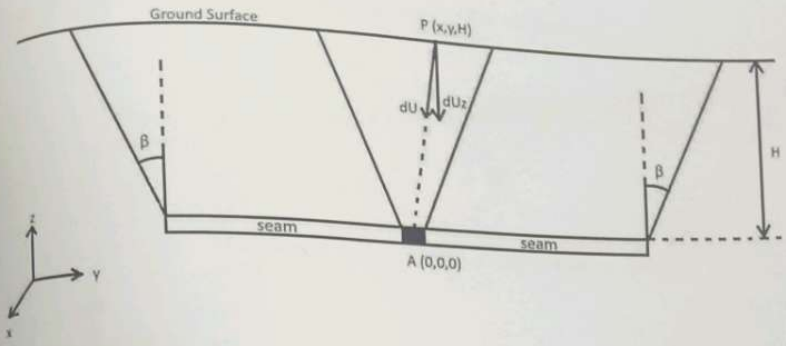


Figure 5

$$dU_z(x,y) = S_{\max} \frac{\exp(-\pi(x^2+y^2)/a^2)}{a^2}\, dx\, dy.$$

Given, $a = H \tan\beta$, and Smax = Maximum subsidence value. Equation 5

Through this dU at P can be calculated due to infinitesimal small excavation at point A and then by that dUx and dUy using equation 2, 3, 4a and 4b.

Now, the seam is divided in many infinitesimal small excavation and dUx, dUy and dUz is calculated at every point on the ground surface due to all infinitesimal small excavation and then they are superposed to get the final Ux, Uy and Uz value at every point.

Now, we will use this similar idea on inclined seam also but in a different way.

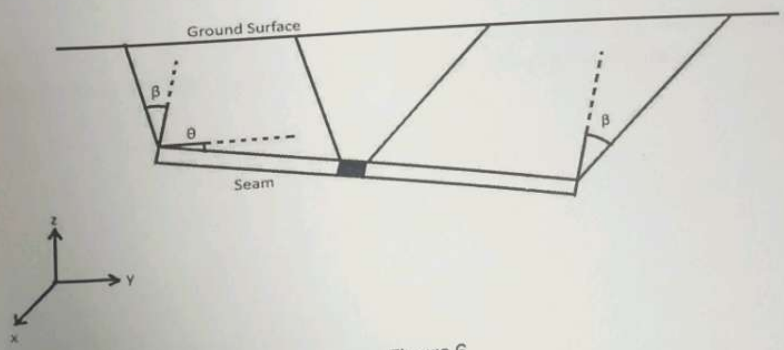


Figure 6

Now, we will transform the above scenario to the below one.

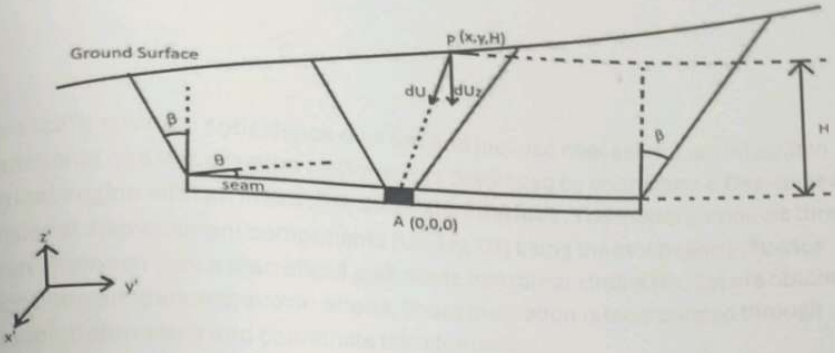


Figure 7

Now, we will apply here the stochastic influence in the same way in which we apply on the flat seam.

$$dU_z(x,y) = S_{\max} \frac{\exp\left(-\pi(x^2+y^2)/a^2\right)}{a^2}\, dx\, dy.$$

Given, $a = H \tan\beta$ and Smax = Maximum subsidence value. Equation 6

Through this dUx' and dUy' at P can be calculated due to infinitesimal small excavation at point A using equation 2, 3, 4a and 4b.

Now, the seam is divided in many infinitesimal small excavation and dUx', dUy' and dUz' is calculated at every point on the ground surface due to all infinitesimal small excavation and then they are superposed to get the final Ux', Uy' and Uz' value at every point.

Now, after this we can easily transform (Ux', Uy', Uz') to (Ux, Uy, Uz) because we have changed the co-ordinate system (x-y-z) in Figure 6 to (x'-y'-z') in Figure 7.

Chapter 3: Computational Implementation

To numerically evaluate subsidence over flat and inclined coal seams, an integrated computational and visualization platform was developed by combining a **C++-based numerical engine** with an interactive **software interface**. The model computes three-dimensional displacement components (Ux, Uy, Uz) using the stochastic influence function approach over a discretized grid, while horizontal strains (Ex, Ey) are obtained through finite difference approximations. Seam inclination is incorporated through orientation parameters and coordinate transformation.

The software replaces traditional file-based input with a graphical interface, allowing users to define field dimensions, grid resolution, seam orientation, reference coordinates, extraction boundaries, and seam properties directly. It supports flexible modeling of complex panel geometries.

Computed results are visualized as color-mapped fields, with additional features such as statistical summaries, interactive value inspection, and user-defined cross-sectional plotting for detailed subsidence analysis.

4.1 Input Structure

The software replaces static input files with a structured graphical input system. Users provide the following parameters through the interface:

- **Field and grid parameters:** field length, field breadth, and grid resolution (dx, dy)
- **Seam orientation and geometry:** direction cosines (ns_x, ns_y, ns_z) and reference coordinates (x_ref, y_ref, z_ref)
- **Seam properties:** subsidence factor (p) and angle of draw (β)
- **Excavation boundaries:** multiple user-defined rectangular regions representing extracted panels

This flexible input architecture allows rapid modification of mining scenarios without recompilation or file editing, significantly improving usability and applicability for real-time subsidence analysis.

4.2 Grid Generation

The computational field is discretized into a **rectangular mesh**. The number of elements in the x and y directions are computed as:

$$I = \left\lceil \frac{\text{Field Length}}{dx} \right\rceil, \quad J = \left\lceil \frac{\text{Field Breadth}}{dy} \right\rceil$$

Equation 7a and 7b

Each cell represents a surface computation point (Px,Py,Pz), where:

$$Px = m \cdot dx + \frac{dx}{2}, \quad Py = n \cdot dy + \frac{dy}{2}, \quad Pz = 0$$

Equation 8a, 8b and 8c

A **binary seam map** is constructed to mark extraction zones. Cells with 1 denote minedout grid elements and are used as influence sources during subsidence calculation.

4.3 Subsidence Calculation Routine

The core computation is carried out in the function **`Calculate_Subsidence`**, which performs superposition of influence from each extracted element to every surface point. For each mined element:

- Its coordinates (xa, ya, za) are computed based on seam orientation.
- The **distance to each surface point** is decomposed into:
  - **normal component (Depth)** controlling influence strength
  - **horizontal radial component (r)** controlling lateral spread

The influence radius Rf is defined as:

$$Rf = \text{Depth} \cdot \tan(\beta)$$

Equation 9

and the maximum subsidence at the ground surface due to a unit element is computed using:

$$S_{\max} = \frac{p}{n_s^z}$$

Equation 10

The influence function integrated over the grid cell area results in incremental displacement:

$$\phi = S_{\max} \cdot \frac{e^{-\pi\left(\frac{r^2}{Rf^2}\right)}}{Rf^2} \cdot (dx\, dy)$$

Equation 11

The 3D displacement vector is obtained by resolving the incremental movement along the direction connecting the mined element to the surface point. This ensures that the model naturally accommodates **vertical as well as horizontal displacement components**, thereby maintaining theoretical consistency with physical strata movement.

The computed displacements are cumulatively stored in displacement matrices:

Ux (m,n), Uy (m,n), Uz (m,n)

4.4 Strain Calculation

After surface displacements are computed, **horizontal strain fields** are evaluated numerically using centered finite difference operators:

$$E_x(i,j) = \frac{U_x(i+1,j) - U_x(i-1,j)}{2\,dx}$$
$$E_y(i,j) = \frac{U_y(i,j+1) - U_y(i,j-1)}{2\,dy}$$

Equation 12a and 12b

These strain components are critical for infrastructure damage assessments, as high curvature and tensile strain magnitudes correlate strongly with structural distress.

4.5 Output and Visualization

The computed displacement and strain fields are directly visualized within the developed software interface, eliminating dependence on external visualization tools. The results include full displacement components (Ux, Uy, Uz) and horizontal strain components (Ex, Ey), displayed as color-mapped spatial distributions over the computational grid.

The platform supports interactive visualization features such as dynamic field selection, customizable color maps, real-time statistical summaries, and cursor-based value inspection. Additionally, users can generate cross-sectional plots along user-defined lines to analyze subsidence profiles in detail.

These capabilities enable clear interpretation of subsidence behavior, including trough formation, displacement trends, and strain distribution, for both flat and inclined seam conditions.

4.6 Computational Performance

The algorithm scales with:

$$O(Ns \times Ng)$$

where:

- Ns : number of mined grid elements
- Ng : number of surface grid points

Even for high-resolution meshes (10,000+ points), execution time remains reasonable due to:

- Loop optimization
- Cache-friendly vector containers
- Avoidance of expensive memory reallocation

Chapter 4: Case Study: Adriyala Mine

6.1 Description of the Adriyala Mine

- Mine in Adrial, Telangana, India.
- Long Wall Panel-1 Dimensions: Length = 2333.00 m, Width = 250.00 m
- Seam Thickness = 6.50 m to 7.00 m
- Seam Gradient = 1 in 5.5
- Height of Extraction = 3.50 m
- Depth of Panel = 456.68 m (Maximum Depth), 366.12 m (Minimum Depth)
- Monitoring stations: number, spacing, measurement method
- Long Wall Panel started on 15.10.2014 (Observation Period Started)
- Long Wall Panel Completed on 29.12.2016
- Long Wall Panel Sealed on 27.10.2017 (Observation Period Ended)

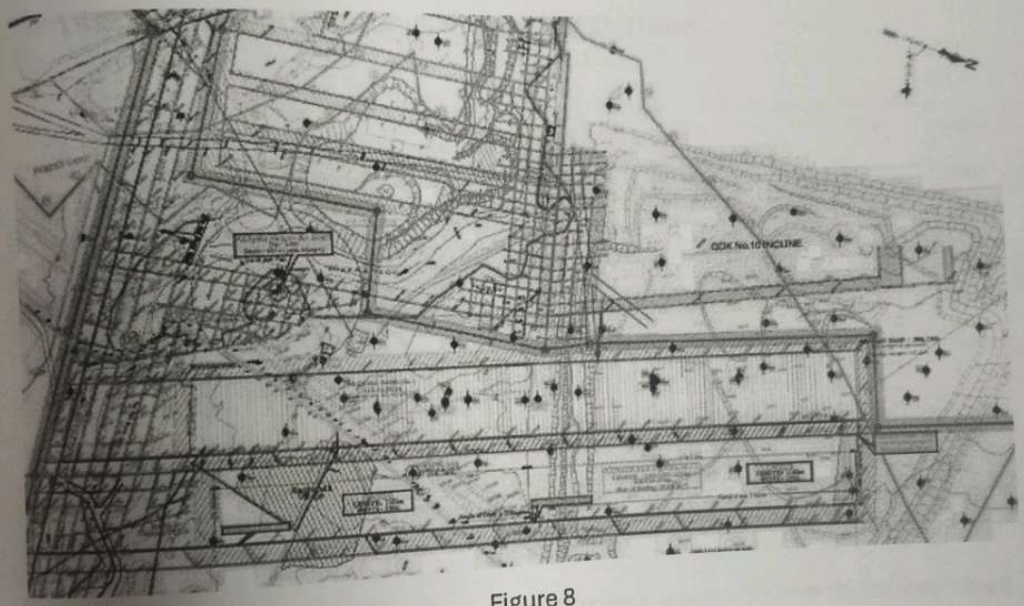


Figure 8

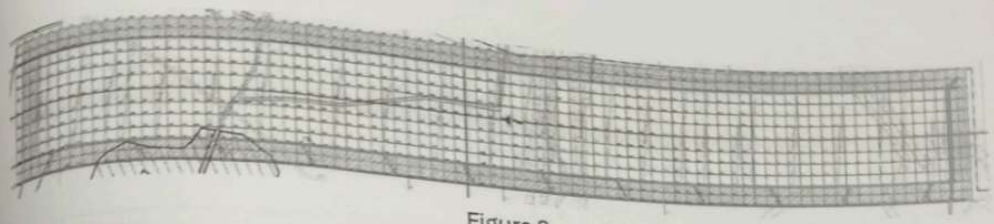


Figure 9

6.2 Input Parameters Used in Numerical Model

- Field Length and Field Breadth = 2310.00 m X 470.00 m
- Grid resolution (dx, dy) = (10, 10)
- Seam orientation parameters (ns_x ,ns_y ,ns_z) = (-0.0186, -0.1779, 0.9838)
- Reference coordinate on the seam plane = (0.00, 120.00, -456.68)
- Spatial coordinates defining extracted regions (closed boundaries) x1 = 0.00 m to x2 = 2310.00 m and y1 = 120.00 m to y2 = 370.00 m
- p: maximum subsidence factor = 1.3 m (By Hit & trial)
- β: influence angle = 20°

6.3 Actual Subsidence Data vs Model Predictions

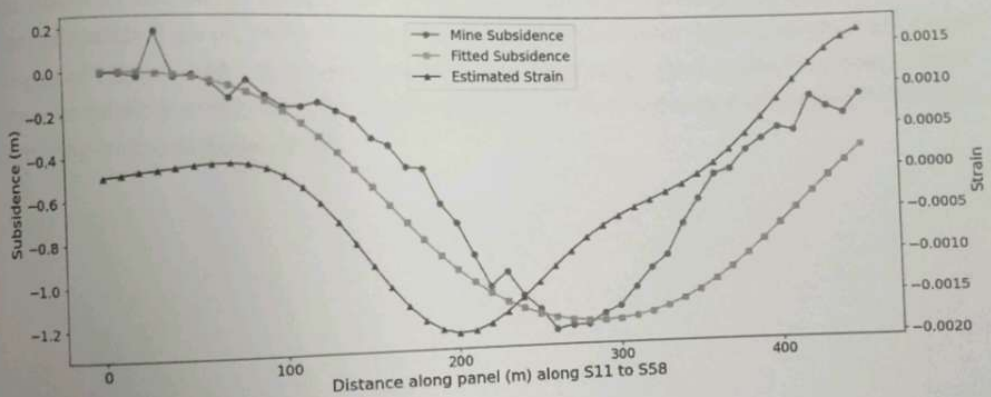


Figure 10

The developed stochastic influence function for inclined seams is fitted to the measured subsidence data along monitoring line S11–S58 to estimate the model parameters Smax and β. The comparison between the fitted and observed subsidence profiles is presented in Fig. 10.

The estimated values of the model parameters are Smax=1.3 and β=20°. To evaluate the predictive capability of the model, the inclined-seam IFM is applied to an independent dataset along monitoring line S61–S99 using these parameters. The predicted subsidence profile for this testing region is presented in Fig. 11.

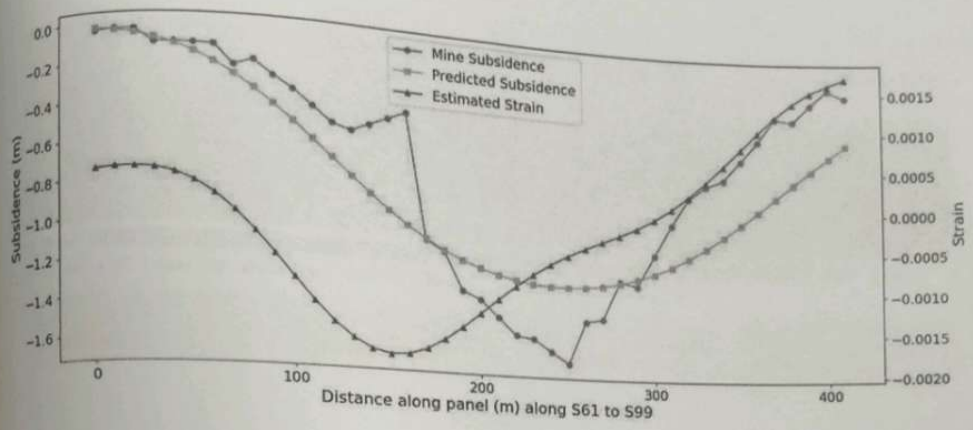


Figure 11

As shown in Fig. 10 and 11, the developed IFM for inclined seams successfully captures the subsidence behavior of the studied panel. The RMSE for the training dataset is approximately 0.25 m, while that for the testing dataset is about 0.26 m, within subsidence ranges of 1.25 m and 1.6 m, respectively. These results indicate that the proposed model provides reliable and consistent predictions of both the magnitude and spatial distribution of mining-induced subsidence.

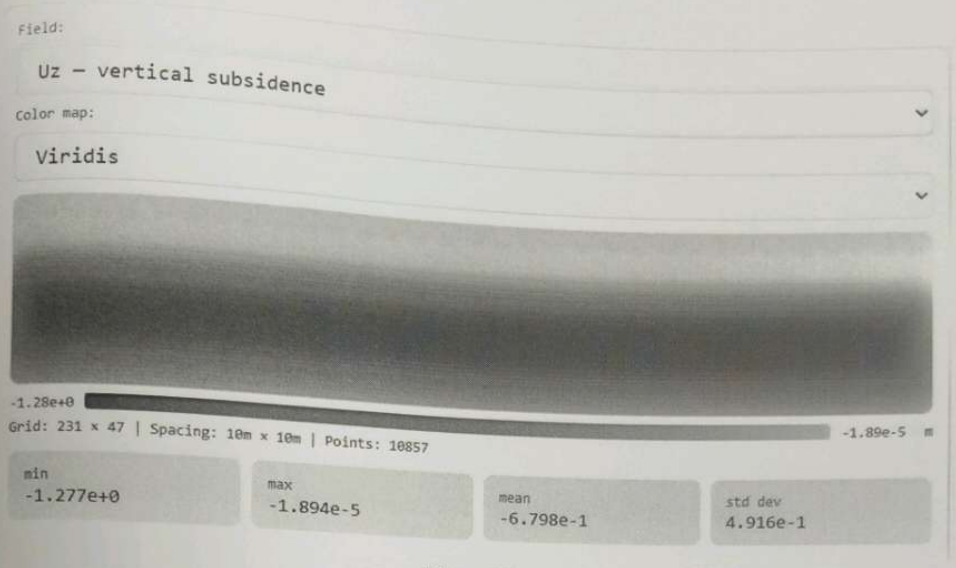


Figure 12

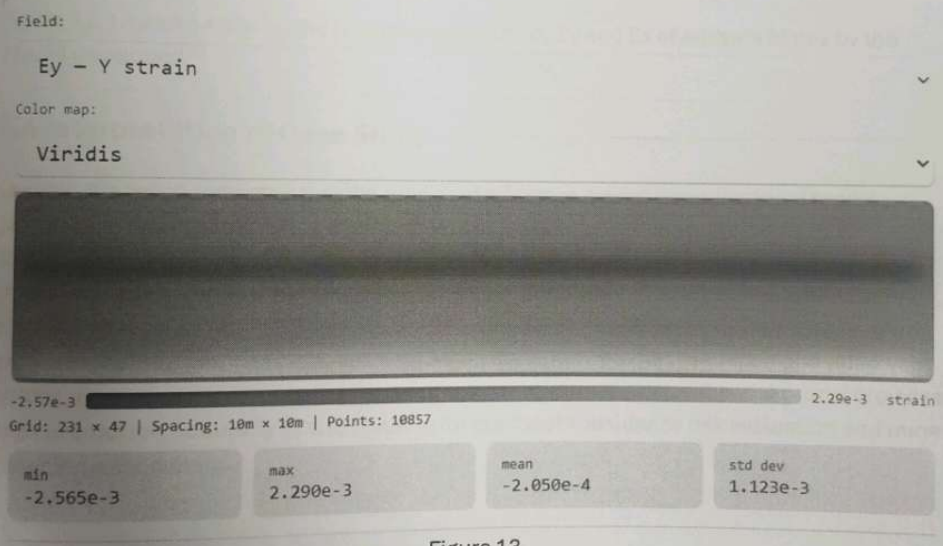


Figure 13

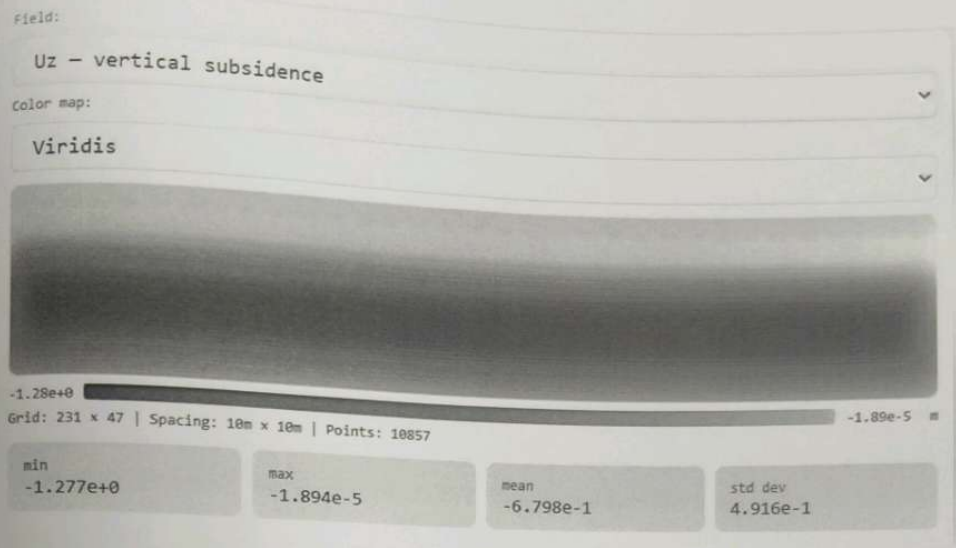


Figure 12

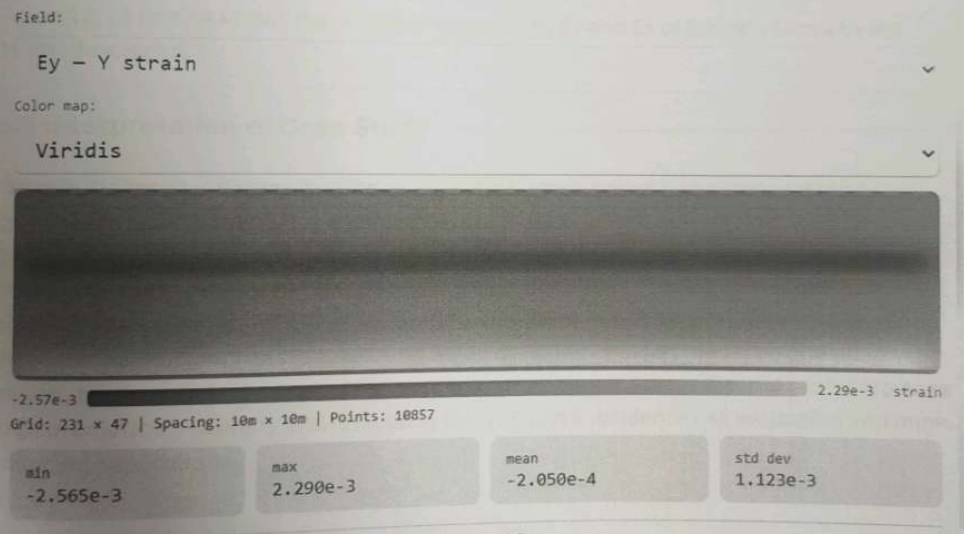


Figure 13

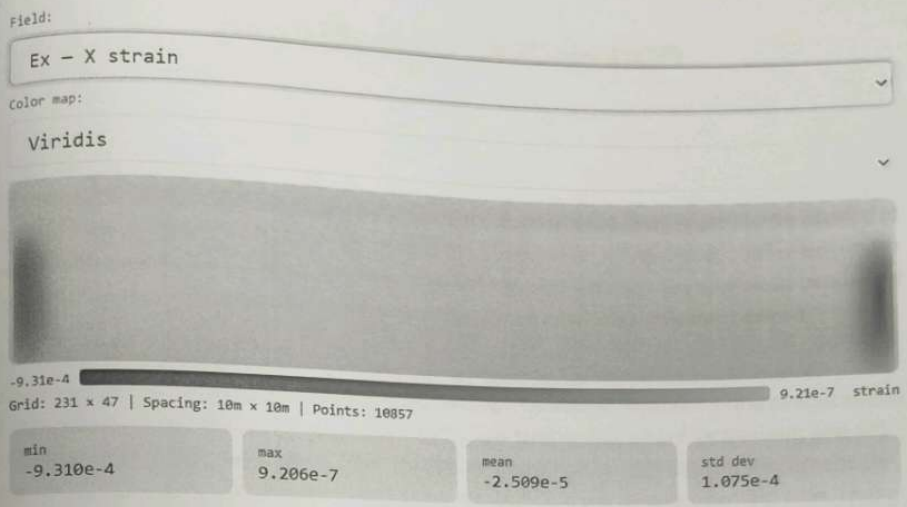


Figure 14

Figure 12,13 and 14 shows the predicted value of Uz, Ey and Ex of Adriyala Mines by the Model developed.

6.4 Interpretation of Case Study

Overall, the result indicates that the proposed influence-based computational model performs reliably in predicting mining-induced surface subsidence over the Adriyala longwall panel. The predicted curves closely follow the measured subsidence envelopes if noise is removed along both monitoring lines, successfully capturing the **location of maximum settlement, width of the trough, and asymmetric deformation trend caused by seam inclination**. The comparatively small deviation between predicted and observed values, particularly in the central region of the panel, demonstrates that the model provides **high prediction accuracy** and is suitable for practical subsidence risk evaluation and mine planning applications.

Results and Discussion

The computational subsidence prediction model developed in this research was applied to the Adriyala longwall panel to evaluate its performance for an **inclined coal seam under real field conditions**. The results were assessed through a comparison between **predicted** and **field-measured (actual)** subsidence along multiple surface monitoring lines.

The predicted subsidence profiles successfully reproduced the **global deformation characteristics** observed in the field. In particular, the model captured (i) the overall shape of the subsidence basin, (ii) the location and magnitude of maximum settlement, and (iii) the **asymmetry of the trough caused by seam inclination** — with steeper settlement on the shallower up-dip side and a more gradual recovery on the deeper downdip side. These patterns were consistent across all monitoring sections.

Although the predicted curves were generally smooth, the actual measurements exhibited **localized spikes and dips**, likely arising from abrupt lithological variations, joint activation, caving irregularities, or surface measurement uncertainties. The influence function formulation assumes a continuum deformation behaviour and therefore does not explicitly simulate sudden point-level fluctuations. However, the **deviation between the actual and predicted data remained small over most of the subsidence trough**, indicating that the dominant mechanisms of strata movement were accurately captured.

The agreement between measured and predicted subsidence — particularly in terms of subsidence magnitude and basin width — demonstrates that the proposed extension of the Influence Function Method for inclined seams provides a **credible and computationally efficient representation of field deformation behaviour**. The ability of the model to generate realistic subsidence troughs without requiring multiple calibration constants further reinforces its suitability for **mine design, subsidence-risk evaluation, and pre-extraction prediction studies**.

Conclusion

This research presented a comprehensive investigation into the prediction of mining induced surface subsidence using the **Influence Function Method (IFM)** and its stochastic extension, with a new mathematical formulation developed to model subsidence over **inclined coal seams**. The work progressed from theoretical study to numerical implementation, culminating in validation against real field subsidence data.

The key achievements of this research are as follows:

1. A detailed theoretical framework for influence-based subsidence modelling was studied, highlighting the limitations of the classical IFM when applied to non-flat seams.
2. The **Stochastic Influence Function Method (SIFM)** was implemented to incorporate variability in overburden behaviour while preserving analytical simplicity.
3. A novel extension of the influence function concept was formulated to predict subsidence for **inclined seams**, by modelling seam depth as a spatial function along the dip direction rather than modifying the influence function itself.
4. A complete **C++ computational tool** was developed to evaluate vertical and horizontal displacement and strain over the ground surface using discretized influence superposition and a **use friendly software** is developed to visualize the data.
5. The model was validated using actual monitoring data from the **Adriyala longwall panel**, showing strong agreement between predicted and observed subsidence behaviour and accurately capturing the asymmetric trough caused by seam dip.

It is important to note that the present model is primarily applicable to **moderately inclined seams**—typically up to **about 12°**—where the geometric distortion of the influence field remains within a tractable range. **Steeply inclined seams** require further enhancements beyond the current formulation.

Overall, the research delivers a **theoretically consistent, computationally efficient, and practically useful approach** for subsidence prediction in longwall mining, with strong potential for integration into mine planning and subsidence risk management frameworks.

Learning Outcomes and Future Scope

Learning Outcomes

Through the course of this research, the following technical insights were achieved:

- Deep understanding of subsidence mechanisms and strata behaviour during longwall mining.
- Expertise in analytical subsidence modelling using **influence functions and probability-based deformation fields**.
- Practical experience in converting a theoretical geomechanics model into a **numerical C++ implementation capable of handling real-scale mine data**.
- Understanding of challenges involved in validating computational models against **field-measured subsidence**.
- Awareness of how geological complexity, seam inclination and panel geometry influence surface deformation.

Future Scope

Although the proposed model provides reliable subsidence prediction for moderately inclined seams, several opportunities exist to extend this research further. Some possible directions are:

1. **Prediction for steeply inclined seams (>12°):**
   The present formulation assumes moderate seam inclination, where the deformation still propagates in a predictable geometric manner. For steep inclinations, the overburden movement becomes highly asymmetric and nonlinear. Extending the mathematical model to handle **steep dips** will significantly enhance its applicability in structurally disturbed coalfields.
2. **Multiple-seam and stacked-seam subsidence:**
   Many Indian and international coal basins involve exploitation of multiple seams at different depths. Coal extraction in multiple horizons results in **superimposed and interaction-driven subsidence troughs** which cannot be captured by single-seam influence functions. Extending the model to simulate **seam-to-seam interactions and cumulative subsidence effects** is an important future direction.

3. **Dynamic subsidence / time-dependent deformation:**
   The current model calculates final subsidence after full extraction. Introducing **progressive deformation modelling** or **creep-based time evolution** can help forecast subsidence during panel advancement, which is crucial for infrastructure monitoring and mitigation planning.
4. **Geologically heterogeneous formations:**
   The present formulation assumes continuum behaviour of the overburden. Incorporating layer-wise rock stiffness or lithological interfaces can improve prediction of **localized subsidence spikes** observed in field data.
5. **Integration with risk and damage assessment tools:**
   The developed model may be coupled with civil infrastructure vulnerability models to evaluate **potential surface damage**, which will enhance its usability for **land use planning and environmental compliance**.

References

1. Chen, Y., Zha, J. and Wang, L., 2025. Adaptive dynamic prediction model of mining subsidence aided by measured data. Scientific Reports, 15(1), pp.1-15.
2. Bahuguna, P.P., Singh, B., Srivastava, A.M.C. and Saxena, N.C., 1993. Semi-empirical method for calculation of maximum subsidence in coal mines. Geotechnical & Geological Engineering, 11(4), pp.249-261.
3. Luo, Y., 2015. An improved influence function method for predicting subsidence caused by longwall mining operations in inclined coal seams. International Journal of Coal Science & Technology, 2(3), pp.163-169.
4. Xu, N., Kulatilake, P.H., Tian, H., Wu, X., Nan, Y. and Wei, T., 2013. Surface subsidence prediction for the WUTONG mine using a 3-D finite difference method. Computers and Geotechnics, 48, pp.134-145.
5. Ren, G., Li, J. and Buckeridge, J., 2010. Calculation of mining subsidence and ground principal strains using generalised influence function method. Mining Technology, 119(1), pp.34-41.
6. Deb, D. and Verma, A.K., 2025. Fundamentals and applications of rock mechanics. PHI Learning Pvt. Ltd..